

Software Requirements Specification (SRS)

**Project Title: Citizen Complaint Management System
(CCMS)**

Domain: Politics and Governance

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Revision History

Name	Date	Reason For Changes	Version
APSS Fernando	02.06.2025	Initial draft of SRS document	0.1
APSS Fernando	04.06.2025	Added use case diagram	0.2
APSS Fernando	06.06.2025	Final review before submission	1.0

1. Introduction

1.1 Purpose

The Citizen Complaint Management System (CCMS) enables citizens to report governance-related issues such as corruption, service delivery delays, and policy violations directly to local government officials. It aims to promote transparency, accountability, and civic engagement in politics and governance.

1.2 Document Conventions

- Requirements labeled as REQ#
- Priorities: High (H), Medium (M), Low (L)

1.3 Intended Audience and Reading Suggestions

Audience

Developers, Testers, Project Managers, Users/Admins

Reading Suggestions

For quick overview: Read Section 1 (Introduction) and Section 2 (Overall Description).

For technical details: Refer to Sections 3–5.

For governance compliance: Review Section 5.2 (Security Requirements).

1.4 Product Scope

Citizens need effective ways to report political and governance concerns including:

- Election administration issues
- Public official misconduct
- Policy implementation failures
- Government service delays

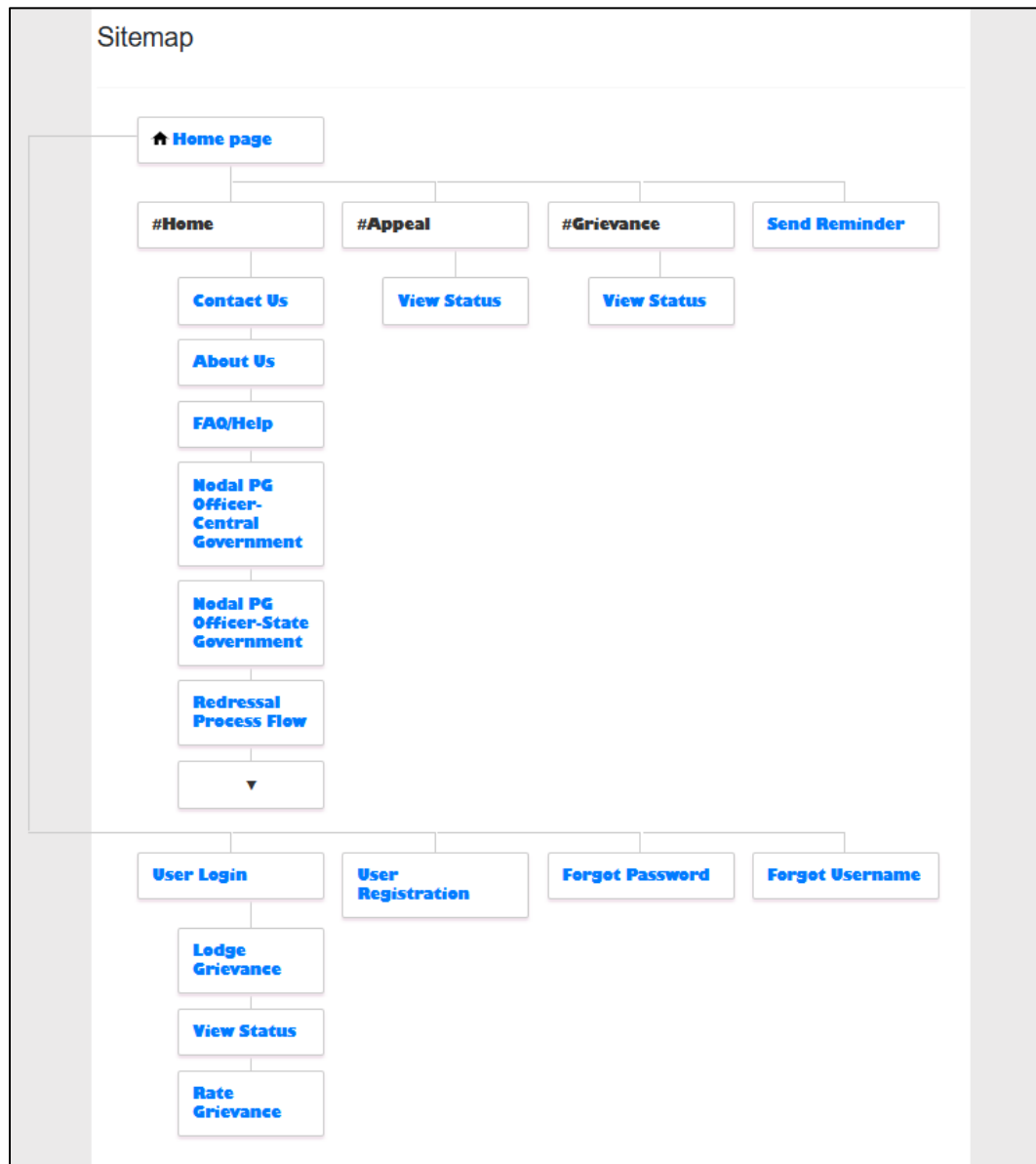
The Citizen Complaint Management System (CCMS) will provide:

- A secure platform for political grievance reporting
- Transparent tracking of complaint resolution
- Direct channels to accountability bodies
- Public dashboards showing resolution rates

By implementing CCMS, the government will significantly enhance political accountability while fostering greater citizen participation in governance processes. This digital solution will modernize complaint management, reduce bureaucratic barriers, and strengthen democratic oversight mechanisms across all levels of government.

1.5 References

- IEEE SRS Template
- India's CPGRAMS (Centralized Public Grievance Redress and Monitoring System)



<https://pgportal.gov.in/> - CPGRAMS

2. Overall Description

2.1 Product Perspective

The CCMS is a new digital platform consisting of a web application and a mobile app, designed to enhance transparency and civic engagement in governance. The system will be integrated with government databases and communication channels (such as email and SMS), and may also support social media alerts in the future to improve public accountability.

2.2 Product Functions

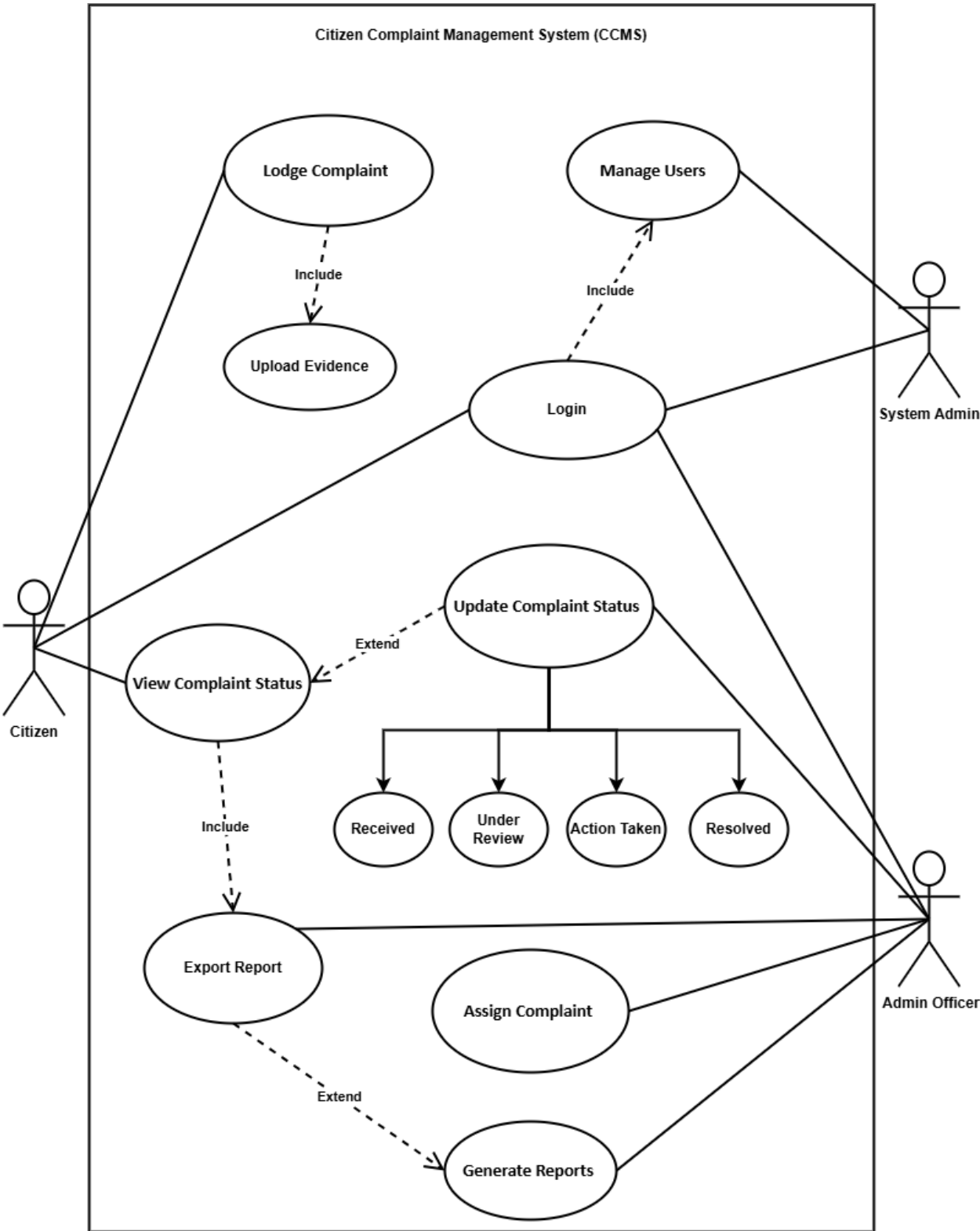
The system shall provide the following functions:

- User Registration and Login
- Complaint Submission
- Complaint Tracking
- Administrator Dashboard for complaint management
- Report generation and analytics

2.3 User Classes and Characteristics

User Type	Who They Are	Technical Skill	Main Concerns or Tasks
Citizen	Regular people using the system to report issues or complaints.	Low (Non-technical)	Wants the system to be easy to use. Cares about results, fairness, and transparency in how complaints are handled.
Admin Officer	Government staff who review, manage, and update complaint records.	Medium (Technical)	Uses the system often. Needs tools to view, assign, and respond to complaints quickly and correctly.
System Admin	IT expert who maintains and troubleshoots the system.	High (Highly technical)	Ensures the system works smoothly, manages user accounts, security, backups, and performance.

Use Case Diagram:



2.4 Operating Environment

- Web Server : Apache or Nginx
- Backend Language : PHP
- Database : MySQL
- Supported Browsers : Chrome, Firefox, Edge
- Access via Web browsers and Mobile apps

2.5 Assumptions and Dependencies

2.5.1 Assumptions

- The system will initially support only the English language. Support for Sinhala and Tamil may be added later.
- Citizens will have access to the internet or mobile data to use the system via a web or mobile app.
- Citizens will provide accurate and complete information while submitting complaints.
- All users must register using a valid National ID number to access the platform.
- Government officers and administrators will have the necessary digital literacy to use the system.
- The SMS and email services used to send notifications will be available and stable.
- The complaint handling workflow will depend on manual review and assignment by admins (i.e., auto-routing will be basic or limited in initial versions).

2.5.2 Dependencies

- Integration with the National Identity Verification API is required for user authentication.
- The system depends on third-party services for SMS and email delivery.
- Hosting and deployment will rely on government-approved cloud infrastructure or internal data centers.
- Any changes in government complaint-handling policies or data privacy regulations may affect the requirements or operation of the system.

3. External Interface Requirements

3.1 User Interfaces

- The system will provide a responsive and accessible user interface, optimized for both desktop and mobile devices.
- Interface will support multiple languages.
- Complaint submission forms will include proper input validation, tooltips, and clear error messages.
- Common UI components will include:
 - A help button or FAQ section
 - A logout option for all user types
 - Navigation bar for Citizens, Officers, and Admins based on their roles

3.2 Hardware Interfaces

- No specialized hardware is required to operate the system.
- The system will run on standard desktop/laptop computers, tablets, and smartphones with internet connectivity.

3.3 Software Interfaces

Integration with:

- Email APIs to send automated updates and complaint receipts
- SMS APIs for OTP verification and status alerts
- National ID Verification API for validating user identity during registration
- Optional: Social media integration (e.g., Twitter/Facebook) to publicly show resolved issues or raise awareness

3.4 Communications Interfaces

- The system will use HTTPS (SSL/TLS encryption) to secure all data transfers between users and the server.
- A RESTful API will be used to connect the web application to the mobile app.
- All communication protocols will adhere to government security standards for public data handling.

4. Functional Requirements

4.1 User Registration and Login

REQ-1 (H):

As a citizen,

"I can register on the platform using my email or phone number and a secure password, so that I can access personalized services and submit complaints securely".

REQ-2 (H):

As a citizen,

"I can reset my password using a recovery link sent to my email or phone number, so that I can regain access to my account without needing admin help".

4.2 Complaint Submission

REQ-3 (H):

As a citizen,

"I can fill in a form to submit a complaint, including text, selecting a location, and uploading images or videos, so that the responsible authority receives complete and relevant information to take action".

REQ-4 (H):

As a citizen,

"I receive an automated confirmation message (email or SMS) after submitting a complaint, so that I have proof that my complaint was successfully registered".

4.3 Complaint Tracking

REQ-5 (H):

As a citizen,

"I can log in and view the real-time status of my submitted complaints, including updates or remarks from officials, so that I remain informed about the progress and actions taken on my issue".

4.4 Administrator Dashboard

REQ-6 (H):

As an admin officer,

'I can view and filter submitted complaints by category, date, location, or status and assign them to relevant departments, so that complaints are distributed efficiently and resolved faster'.

REQ-7 (H):

As an admin officer,

"I can update the status of complaints (e.g., under review, resolved) and add remarks or feedback, so that citizens are notified of progress and outcomes".

4.5 Report Generation

REQ-8 (H):

As an admin officer,

"I can generate visual and textual reports showing complaint trends based on type, location, and resolution times, so that I can evaluate governance performance and identify recurring issues".

REQ-9 (M):

As an admin officer,

"I can export generated reports in PDF and CSV formats, so that I can share them with decision-makers or archive them for future analysis".

5. Non-Functional Requirements

5.1 Performance

REQ-10 (H):

As a user,

"I want each page of the system to load and respond within 2 seconds even during normal use, so that I have a fast and seamless experience without frustration".

REQ-11 (H):

As an admin,

"I want the platform to handle up to 1000 concurrent users without slowing down, so that the system remains functional during public campaigns or crises".

5.2 Security

REQ-12 (H):

As a user,

"I want my session to automatically log out after 10 minutes of inactivity, so that my account is protected from unauthorized access".

REQ-13 (H):

As a user,

"I want my passwords to be stored using strong encryption techniques like industry-standard hashing algorithms, so that even in the case of a data breach, my credentials are safe".

REQ-14 (H):

As a user,

"I want all my interactions with the system (logins, submissions, viewing data) to happen over HTTPS, so that my personal data is not intercepted by attackers".

5.3 Usability

REQ-15 (H):

As a citizen,

"I want the system interface to be mobile-friendly, with readable text and easy-to-use buttons, so that I can quickly report issues on the go from my phone or tablet".

REQ-16 (M):

As an admin,

"I want a clear and organized interface with tooltips and minimal learning curve, so that I can manage complaints without requiring extra training".

5.4 Reliability & Availability

REQ-17 (H):

As a citizen,

"I want the system to be operational and accessible at least 99% of the time, so that I can report issues or check updates whenever needed".

REQ-18 (H):

As an admin,

"I want the system to perform automatic daily backups and have failover mechanisms, so that no data is lost and services are restored quickly during outages".

5.5 Compliance & Localization

REQ-19 (H):

As a project manager,

"I want the platform to comply with local laws such as data protection and digital governance standards, so that the system remains legally approved and trustworthy for public use".

REQ-20 (M):

As a citizen,

"I want to use the platform in my preferred local language, so that I can understand and interact with all features easily, regardless of my language skills".

REQ-21 (M):

As an admin officer,

"I want the system to automatically escalate unresolved complaints to higher authorities after 30 days, so that no complaint gets stuck indefinitely without resolution."

6. Other Requirements

Appendix A: Glossary

Term / Acronym	Definition
CCMS	Citizen Complaint Management System – The proposed web and mobile platform for citizens to lodge and track complaints to government authorities.
Citizen	A non-technical user who submits complaints and expects resolution and transparency from government bodies.
Complaint	A formal submission by a citizen regarding issues or grievances with government services.
Complaint ID	A system-generated unique identifier assigned to each complaint.
Status	The current stage of a complaint (e.g., Submitted, In Progress, Resolved, Rejected).
Admin Officer	A government official responsible for reviewing and responding to citizen complaints within their assigned department or jurisdiction.
System Admin	A technical user responsible for managing the CCMS infrastructure, user accounts, and system settings.
Dashboard	A visual interface displaying metrics such as the number of complaints, their status, and user activity.
Feedback	Citizen input or rating given after a complaint has been resolved.
OTP	One-Time Password – A temporary code sent via SMS/email to verify user identity during login.
Jurisdiction	The specific geographical or administrative area for which a government officer is responsible.
GIS	Geographic Information System – May be used to map complaint locations.
Notification	Alerts (via email, SMS, or app) sent to users to update them on the status of their complaint.
API	Application Programming Interface – Used to integrate CCMS with external services such as government databases or SMS gateways.
Mobile App	A smartphone-based interface for CCMS users to file, track, and manage complaints.
Web Portal	The browser-accessible platform for users to interact with the CCMS system.
SMS Gateway	A service that allows the system to send text messages (e.g., OTPs or complaint updates) to users.
Authentication	The process of verifying a user's identity, typically through a username/password or OTP.
Role-based Access	A method of controlling what actions a user can perform based on their assigned role (Citizen, Admin Officer, System Admin).

Appendix B: To Be Determined List

- TBD-1: GIS Mapping integration for geolocation tagging of complaints
- TBD-2: Audit trail/logs for complaint activity
- TBD-3: Integration with mobile push notification services
- TBD-4: Scheduled performance tests and up-time monitoring tools
